USB Device Monitor

An Android app that monitors USB device attach/detach events for the **Zebra RFD40 RFID Sled** and displays live status on a modern Material UI.

Features

- **Live status indicator** — large green circle (RFD40 Awake) when the device is connected; gray circle (Sleep Mode) when disconnected
- **Device details** — shows Device Count, Vendor ID (VID `1504`), Product ID (PID), and device path on connect
- **USB transport & power monitoring** — decodes the active USB mode (Power-Only / Charging vs. File Transfer / Debug) from `USB_STATE` and power connect/disconnect broadcasts
- **Interface-change counters** — running totals of USB **attach**, **detach**, and combined **total** events for the current session
- **Event log** — timestamped scrollable log of all USB attach/detach, transport, and power events
- **Wake hint** — prompts user to *"Hold RFD40 trigger to wake up RFID Reader"* when in sleep mode
- **Toast notifications** on each attach/detach event

Screenshots

Awake (Connected)	Sleep Mode (Disconnected)
Green circle, icon, VID/PID populated	Gray circle, icon, wake instruction shown

Requirements

- Android **API 34+** (Android 14)
- Zebra RFD40 RFID Sled (VID `1504`, PID `5889`)
- USB Host mode enabled on the Android device

Project Structure

```
app/src/main/
├── java/com/zebra/rfid/usb/
│   └── MainActivity.java          # USB BroadcastReceiver + UI logic
├── res/
│   ├── layout/activity_main.xml   # Material card-based UI
│   └── drawable/
│       ├── shape_circle_awake.xml # Green status circle
│       └── shape_circle_sleep.xml # Gray status circle
build_run.sh                       # Clean → Build → Install → Launch script
```

Build & Run

One-command script (requires connected device)

```
./build_run.sh
```

This script: 1. Checks for a connected ADB device/emulator 2. Runs `./gradlew clean` 3. Runs `./gradlew assembleDebug` 4. Installs the APK via `adb install -r` 5. Launches `MainActivity` 6. Streams `MYUSB` logcat output

Manual

```
./gradlew clean assembleDebug
adb install -r app/build/outputs/apk/debug/app-debug.apk
adb shell am start -n com.zebra.rfid.usb/.MainActivity
```

USB Event Handling

The app registers **two** runtime `BroadcastReceivers`.

`mUsbReceiver` — host attach/detach:

Intent Action	Meaning
`USB_DEVICE_ATTACHED`	RFD40 plugged in — green circle, populate VID/PID
`USB_DEVICE_DETACHED`	RFD40 unplugged — gray circle, clear device info

`mBatteryReceiver` — transport mode + power state:

Intent Action	Meaning
`USB_STATE`	Decode transport mode via `updateUsbModeFromIntent()` (`connected`, `configured`, `mtp`, `ptp`, `mass_storage`, `adb`, `rndis`)
`ACTION_POWER_CONNECTED`	Cable connected — classify as *Power-Only (Charging)* or *File Transfer / Debug*
`ACTION_POWER_DISCONNECTED`	Cable removed — reset decoded USB state

Mode determination: explicit transport function extras (`mtp` / `ptp` / `mass_storage` / `adb` / `rndis`) take priority; if none are present, a `connected && configured` USB is inferred as a data-capable (file-transfer/debug) link.

See USB_ATTACH_DETACH_DESIGN.md for the full design document with code snippets.

Interface-Change Counters

The app maintains session counters for USB interface changes, shown in the **USB INTERFACE CHANGES** card:

Counter	Increments on	Color
Attach (`mAttachCount`)	`ACTION_USB_DEVICE_ATTACHED`	green
Detach (`mDetachCount`)	`ACTION_USB_DEVICE_DETACHED`	red
Total	`attach + detach` (derived)	blue

Each increment is also appended to the event log (`Attach count = N` / `Detach count = N`). Counters are session-scoped — they start at `0` on launch and are not persisted across activity recreation.

Key Log Tags

```
adb logcat -s MYUSB:D MYUSB_STATE:D
```

```
D  Device List Size: 1
D  Device VID: 1504
D  Device PID: 5889
D  USB action: android.hardware.usb.action.USB_DEVICE_DETACHED
D  USB device detached
D  USB action: android.hardware.usb.action.USB_DEVICE_ATTACHED
D  ACTION_USB_DEVICE_ATTACHED
D  --> Vid =1504
```

USB transport/power events are logged under the `MYUSB_STATE` tag:

```
D USB_STATE[broadcast] connected=true, configured=true, mtp=false, ptp=false, mass_storage=false,
D ACTION_POWER_CONNECTED mode=File Transfer / Debug, connected=true, configured=true, fileTransfe
D  ACTION_POWER_DISCONNECTED
```

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